

FACTORY EQUIPMENT SPECIFICATION — Assembly Line 3

Robot Model: KUKA KR 10 R1100-2 (local variant)

Axis Count: 6

Payload: 10 kg

Reach: 1100 mm

Repeatability: ± 0.03 mm

TORQUE LIMITS (JOINT #3 — GRIPPER)

Nominal operating torque: 60–100 Nm

Warning threshold: 110 Nm

Alarm threshold (E-0211): 120.0 Nm

Hardware cutoff (e-stop): 180 Nm

MAINTENANCE INTERVALS

Lubrication: every 250,000 cycles

Bearing inspection: every 500,000 cycles

Bearing replacement: every 1,200,000 cycles or on wear confirmation

CYCLE COUNTER

As of 2026-05-22 23:00 UTC: 1,204,337

Approaching bearing replacement interval.

ALARM CODES (SUBSET)

E-0101: encoder position error

E-0102: encoder velocity error

E-0201: bearing wear detected

E-0211: torque overshoot (gripper joint) — see runbook E-0211_v3_2026.md

E-0305: collision detection

E-0401: thermal overload